

Linear programming

Linear programming

- *find the conditions that optimize a function given some constraints.*
- *the function and constraints can be expressed as linear equations or inequalities.*

Farmer's problem

How to maximise the profit?

- 110 hectares of land
- Budget of \$ 10.000
- Man-days: 1200
- Entire production can be sold



Variety	Cost (Price\$/Hec)	Profit (Price\$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30

Terminology

- **Objective Function:** The function to be optimised
 - the profit
- **Decision (also termed Design) Variables:** influence the objective
 - hectares with wheat and barley
- **Constraints:** restrict or limit the decision variables.
 - available resources, i.e. area, budget and man-days.
- **Non-negativity restriction:** The decision variables should always be greater than or equal to 0.
 - areas with wheat and barley can only be positive.

Linear problem of the farmer



- **Step 1: Decision variables**
 - The total area for growing Wheat = $X_1 \geq 0$ (in hectares)
 - The total area for growing Barley = $X_2 \geq 0$ (in hectares)
- **Step 2: The objective function**
 - net profit of \$50 for each hectare of Wheat and \$120 for each Barley.
 - Our objective function **to be maximized is:** $f(X_1, X_2) = 50X_1 + 120 X_2$

Step 3: The constraints

1. Budget of \$10,000

$$100 X_1 + 200 X_2 \leq 10000 \Rightarrow X_1 + 2 X_2 \leq 100$$

2. 1200 man-days distributed between the crops

$$10 X_1 + 30 X_2 \leq 1200 \Rightarrow X_1 + 3 X_2 \leq 120$$

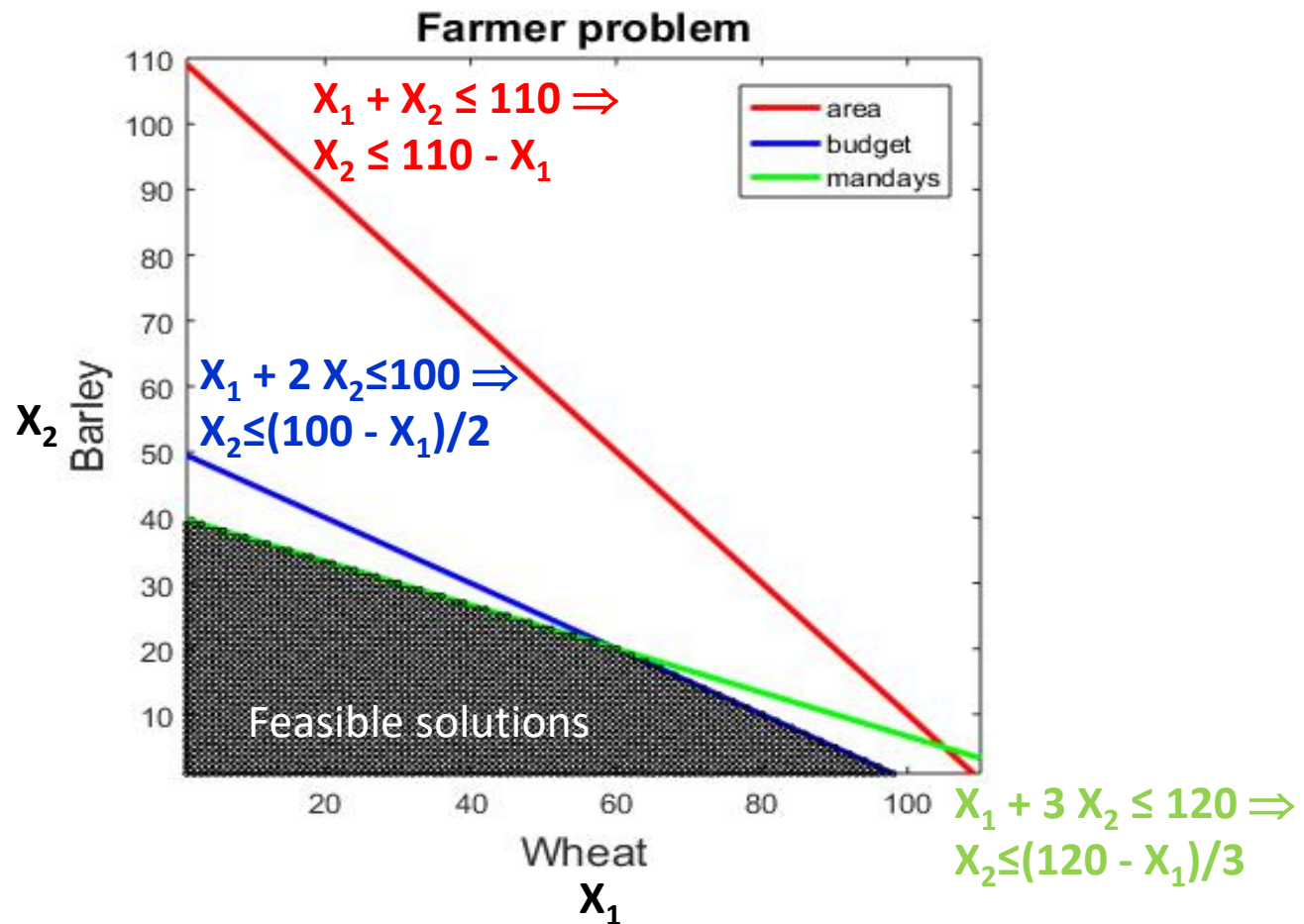
3. Total area is 110 hectares

$$X_1 + X_2 \leq 110$$



Variety	Cost (Price\$/Hec)	Profit (Price\$/Hec)	Man-days/Hec
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Graphical solution



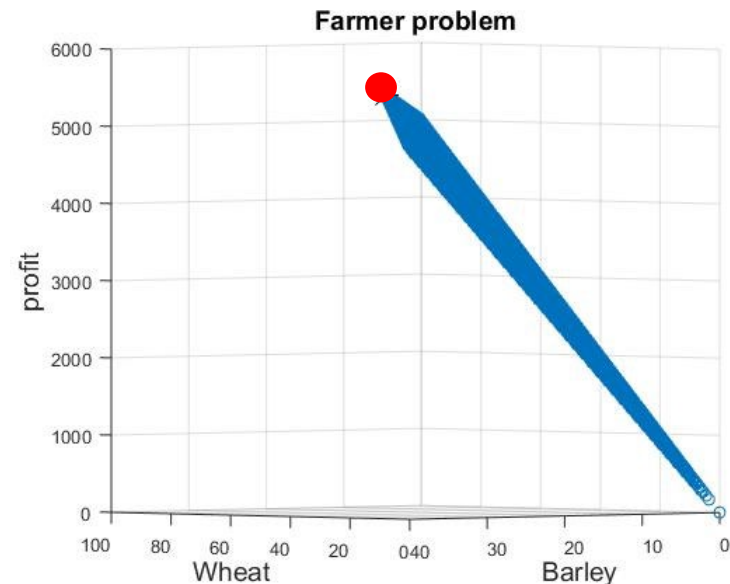
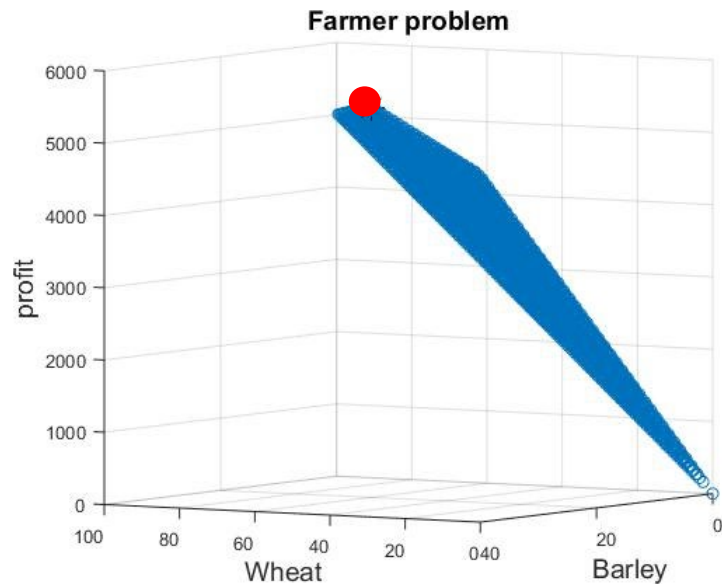


Solution – objective function

60 hectares of wheat

20 hectares of barley

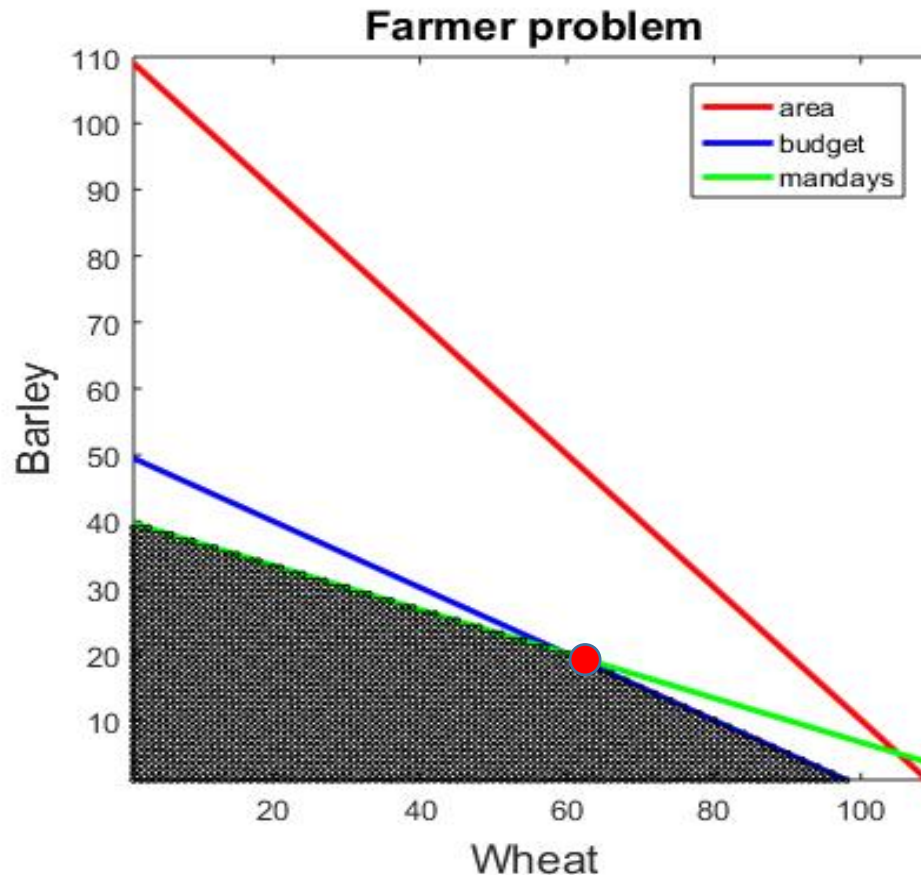
Profit: \$ 5400



Same figure viewed from different angles

Solution

Note: solution will always be at the intersection between lines – possibly extending along a line



Massaging the problem

1. Some inequalities may be stated as

$$X_1 + 2 X_2 \leq 100$$

Others may be stated as

$$X_1 + 3 X_2 \geq 120$$

2. Some constraints may be equalities:

$$X_1 + X_2 = 10$$

3. Optimisation may be maximisation or minimisation of the objective function. The solver we will use minimises.

4. Some x_i may not be ≥ 0

How do we handle these situations?

Massaging the problem

1. Some inequalities may be stated as

$$X_1 + 2 X_2 \leq 100$$

Others may be stated as

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4. Some X_i may not be ≥ 0

Example:

Massage the following linear programming problem

Maximize objective function:

$$x_1 + 2x_2 + 3x_3 + 4x_4 + 5$$

subject to the constraints:

$$4x_1 + 3x_2 + 2x_3 + x_4 \leq 10$$

$$x_1 - x_3 + 2x_4 = 2$$

$$x_1 + x_2 + x_3 + x_4 \geq 1$$

and

$$x_1 \geq 0, x_3 \geq 0, x_4 \geq 0 \text{ - Note no restriction on } x_2$$

How to solve the problem in
MatLab?

Farmer's problem as matrices



$X_1 + 2 X_2 \leq 100$: budget constraint

$X_1 + 3 X_2 \leq 120$: man-day constraint

$X_1 + X_2 \leq 110$: area constraint

The same inequalities in matrix format:

$$\begin{matrix} \mathbf{A} & \mathbf{x} & \leq & \mathbf{b} \end{matrix}$$
$$\begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \leq \begin{pmatrix} 100 \\ 120 \\ 110 \end{pmatrix}$$

The objective function to be maximized: $f(X_1, X_2) = 50 X_1 + 120 X_2$:

$$f^T \cdot x = (50 \quad 120) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

Linprog solves and also does some massaging for us

```
opts = optimoptions(@linprog,'Display','off','Algorithm','dual-  
simplex');
```

```
[x,fval,exitflag] = linprog(f,A,b,Aeq,beq,lb,ub,opts)
```

- $A*x \leq b$ are inequality constraints. $A = []$ and $b = []$ if no inequalities exist.
- $Aeq*x = beq$ are the equality constraints.
- lb and ub are lower and upper bounds on x (remember to set $lb=[0]$ otherwise your problem may be unbounded)
- $Fval$ is the **minimum** value of the objective function
- $Exitflag$ is 1 if a solution is found

Unbounded objective function

There may not be a maximum or minimum of our objective function with the given constraints

Linprog returns an exitflag = -3

No feasible solution

There may not be a solution that fulfils all our constraints, i.e. no feasible solution

Linprog returns an exitflag = -2

Exercises 1

Helping the farmer more

Farmer's wife intervenes



- Optimum solution uses $20 + 60 = 80$ hectares. They have 110.
- They may therefore sell some hectares and increase the budget. The price of land is \$300 per hectare.

Exercise:

Modify the inequalities and find the new optimum conditions

Farmer's father intervenes



Why not plant rye?

Variety	Cost (Price\$/Hec)	Profit (Price\$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30
Rye	150	85	20

Exercise:

How does it look when rye is a possibility?

In this exercise land cannot be sold.

Duality

- Every linear programming problem has a dual problem.
- If the original problem is maximisation, the dual problem is minimisation.

Original problem:

$f^T x$ is to be **maximised**

$$Ax \leq b$$

All elements of $x \geq 0$

Transpose A
Interchange f and b

The dual problem:

$b^T y$ is to be **minimised**

$$A^T y \geq f$$

All elements of $y \geq 0$

$$A^T y \geq f \Rightarrow -A^T y \leq -f$$

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$$-A^T y \leq -f$$

All elements of $y \geq 0$

$[x, fval] = \text{linprog}(-f, A, b, Aeq, beq, lb, ub, opts)$

-f because linprog minimises

$[y, fval] = \text{linprog}(b, -A', -f, Aeq, beq, lb, ub, opts)$

Fval will take the same optimum numerical value

Duality of Farmer's problem



Original problem with selling of land:

$$f^T \cdot x = (50 \quad 120) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \text{ to maximise}$$

$$A x = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \leq \begin{pmatrix} 100 \\ 120 \\ 110 \end{pmatrix} = b$$

All elements of $x \geq 0$

The dual problem:

$$b^T \cdot y = (100 \quad 120 \quad 110) \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \text{ to minimize}$$

$$-A^T y = \begin{pmatrix} -1 & -1 & -1 \\ -2 & -3 & -1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \leq - \begin{pmatrix} 50 \\ 120 \\ 0 \end{pmatrix}$$

All elements of $y \geq 0$

Exercises 2